D03 · Maintenance & Asset Reliability

01 · TPM — Total Productive Maintenance

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1 WHAT IT IS

Total Productive Maintenance (TPM) is the operating philosophy that transfers ownership of routine equipment care from a maintenance department to the operators who run the equipment — supported by eight pillars: autonomous maintenance, planned maintenance, focused improvement, early equipment management, quality maintenance, training and education, safety/health/environment, and TPM in administration. The transfer is not delegation; it is a redesign of the work relationship between operations and maintenance that produces fundamentally different reliability outcomes from traditional siloed models.

2 WHY IT MATTERS

TPM is the most exported and most superficially-implemented maintenance philosophy of the last 40 years — and the gap between the philosophy and its typical implementation has consistently produced the worst pattern in operations: the appearance of TPM without any of its reliability outcomes. **The dominant failure mode is TPM reduced to its visible artifacts — 5S boards, green tags on equipment, operator cleaning checklists — without the operator ownership and skill transfer that make TPM actually work. The plant has the posters; it does not have the philosophy**. The peer-reviewed reference text on TPM implementation in process industries is explicit about this gap **(Suzuki, 1994 — peer-reviewed, "TPM in Process Industries", Productivity Press)**: TPM produces 15–35% MTBF improvement and 40–60% reduction in unplanned downtime **when all eight pillars are implemented with real operator ownership**; it produces nothing measurable when only the autonomous-maintenance pillar is deployed as a 5S program. The asymmetry is structural — the same investment, the same training hours, but radically different outcomes depending on whether the work-relationship redesign happens or whether only the visible artifacts get installed.

3 HOW IT IS DONE

1. **Begin with the autonomous-maintenance pillar (Jishu Hozen) — but understand what genuine autonomous maintenance requires.** The seven steps of Jishu Hozen are not a checklist: initial cleaning (which finds 80% of latent defects), countermeasures for contamination sources, cleaning and lubrication standards, general inspection skills (operator becomes capable of diagnosing equipment), autonomous inspection routines, organization and order, full autonomous maintenance. Each step requires operator certification, not just attendance. Plants that move past step 2 in 6 months and call it TPM produce the 5S theater pattern.
2. **Transfer cleaning-lubrication-inspection (CLI) authority to the operator with formal skill certification — not by edict.** The operator who can clean a pump but cannot tell when it is about to fail has not received TPM training; the operator who can detect bearing degradation by feel and sound has. The skill transfer requires structured training, hands-on practice, written and practical assessment, and ongoing recertification. The plants that produce reliability gains build this skill curriculum; the plants that announce "you now do CLI" produce confused operators and frustrated maintenance technicians.
3. **Implement planned maintenance (the second pillar) in parallel — TPM is not autonomous maintenance alone.** Planned maintenance is what the maintenance organization does after the operator absorbs CLI: predictive analytics, condition monitoring, planned-overhaul scheduling, root-cause failure analysis. The operator handles routine care; maintenance handles the technical work that operators cannot. Plants that treat autonomous maintenance as TPM produce gaps where neither operators nor maintenance own the work; plants that implement both pillars together produce coverage with clear handoffs.
4. **Build focused-improvement teams (the third pillar) chartered against specific reliability losses — not as generic kaizen groups.** Focused improvement under TPM is structured around the 16 equipment losses (the same framework as OEE) with cross-functional teams attacking specific dominant losses. Plants that run focused-improvement teams against unstructured "improvement opportunities" produce activity; plants that charter them against rank-1, rank-2, and rank-3 losses produce reliability gains that compound.
5. **Establish early equipment management (the fourth pillar) so new equipment enters production already TPM-aligned — not as a retrofit afterthought.** The fourth pillar inserts operations and maintenance into the equipment-procurement and

commissioning process, ensuring new equipment is designed for autonomous-maintainability (accessible inspection points, color-coded lubrication, designed-in standard work) from day one. Plants that skip this pillar pay 3–5x more to retrofit autonomous-maintainability into equipment that was not designed for it.

6. **Govern with the TPM steering committee chaired by plant leadership — not delegated to a TPM coordinator with no operational authority.** The eighth pillar (TPM in administration) is operationally the most consequential: TPM either has plant-leadership ownership with the operations director, maintenance director, quality director, and engineering director jointly accountable, or it has a TPM coordinator who runs training and gets ignored when production pressure rises. The governance structure determines whether TPM survives the next operational crisis or becomes the next thing the plant used to do.

WORKED EXAMPLE — ILLUSTRATIVE

A CPG manufacturing site (1 plant, 6 production lines, 220 operators) launches a "TPM program" with consultant support that produces, after 14 months: **5S boards in every cell, color-coded floor markings, green tags on all equipment in good condition, operator cleaning checklists posted**. Reliability outcomes after 14 months: **reactive-to-planned maintenance ratio 78%/22% (unchanged from pre-TPM), MTBF on critical equipment 412 hours (unchanged), unplanned downtime cost $3.6M/year (unchanged)**. Operators escalate to maintenance for 100% of issues, including basic CLI. The 5S boards exist; the philosophy does not. The re-launch redesigns the work relationship: **(1)** formal CLI skill curriculum with 7 certification levels for each operator on the equipment they run; **(2)** structured Jishu Hozen step program with month-by-month targets and operator certification gates; **(3)** planned-maintenance organization restructured to support operator-led routine work and own technical maintenance; **(4)** 3 focused-improvement teams chartered against the top 3 reliability losses identified by equipment data. Results 18 months after re-launch: **reactive-to-planned ratio 78%/22% -> 28%/72%, MTBF 412 hours -> 980 hours (+138%), unplanned downtime cost $3.6M -> $1.3M/year (-64%), maintenance technician hours redirected from firefighting to planned and predictive work**. Same operators, same equipment, same maintenance budget — different work relationship.

IN PRACTICE

Transfer CLI authority with formal skill certification, not by edict

The operator told to perform Cleaning-Lubrication-Inspection without certification training cannot detect early-stage deterioration — the skill is the value, not the task. Plant-led certification matched to the equipment is the structural prerequisite the 8-pillar literature consistently identifies as make-or-break.

Maintenance migrates upward, not out — to predictive, RCM, reliability engineering

The wrong answer is reducing maintenance headcount because "operators are doing the work now" — this produces a death spiral where maintenance resists training operators (their replacements). The right answer: maintenance migrates from reactive firefighting to predictive analytics, planned overhauls, RCFA, and reliability engineering.

Execute Jishu Hozen as the structured 7-step, not as a 5S program with green tags

The 7 steps of autonomous maintenance — initial cleaning, address contamination sources, develop CLI standards, general inspection training, autonomous inspection, organize the workplace, complete autonomous maintenance — are a skill-building sequence, not a tidiness program **(Suzuki, 1994 — peer-reviewed, Productivity Press)**.

Charter focused-improvement teams against rank-1/2/3 reliability losses

The focused-improvement pillar collapses into kaizen-event tourism without ranked loss data. Publish the dollarized reliability-loss Pareto, charter focused-improvement teams only against rank 1/2/3, and require the team output to be a measurable MTBF or downtime change — not a "completed kaizen" artifact.

WHAT YOU WOULD USE

TPM IMPLEMENTATION & TRACKING

IBM Maximo / Infor EAM / SAP PM — CMMS/EAM with TPM workflow — Enterprise asset-management platforms with native support for operator-led work orders, Jishu Hozen step tracking, condition-based maintenance triggers, and reliability KPI rollups; the right fit for multi-plant operations standardizing TPM across sites **(IBM — vendor; Infor — vendor; SAP — vendor)**.

Fiix / UpKeep / eMaint — mid-market CMMS with mobile TPM support — CMMS platforms with strong mobile experience for operator-led inspection and CLI execution, photo capture, and supervisor coaching workflow; the right fit for single-plant operations or operations not on enterprise EAM **(Rockwell/Fiix — vendor; UpKeep — vendor)**.

JIPM (Japan Institute of Plant Maintenance) external certification + internal TPM lead — JIPM provides the canonical TPM Excellence Award framework and certifies plants against the eight-pillar standard; combined with an internal TPM lead (operations background, not maintenance) the structure produces sustained implementation across multi-year timelines **(JIPM — vendor)**.

Cell-level Jishu Hozen boards + supervisor coaching cadence + monthly steering committee — The original TPM approach — physical boards at the cell showing the operator's current Jishu Hozen step status, supervisor weekly review of CLI execution, monthly steering-committee review with operations + maintenance leadership. The lowest-tech option produces the deepest cultural anchor when the cadence and the certification discipline are real.

4 SO WHAT

TPM without genuine operator ownership is a cleaning campaign in Japanese vocabulary — the green tags on the floor do not transfer skills or reduce failures. Begin with autonomous maintenance executed as the structured 7-step Jishu Hozen, transfer CLI authority with formal skill certification not by edict, implement planned maintenance in parallel with operator-led work, charter focused-improvement teams against rank-1/2/3 reliability losses, insert TPM into the equipment-procurement process via early-equipment management, and govern with plant leadership not with a TPM coordinator. The 15–35% MTBF gain TPM is documented to deliver is not a maintenance-program effect; it is a work-relationship effect — and the difference between plants that produce TPM theater and plants that produce TPM outcomes lives entirely in whether the ownership transfer is genuine.

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